

Aagosh Raj Srivastava

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SUMMARY

Aspiring Software Development Engineer dedicated to architecting resilient, high-concurrency backend systems and intelligent workflows. Driven by a strong product mindset to take extreme ownership of complex engineering challenges from initial design through production deployment.

EXPERIENCE

Generative AI Engineer Intern — Finlatics

Jun 2025 – Aug 2025

Remote

- Engineered scalable REST APIs in Python for asynchronous streaming services, cutting Time-To-First-Token (TTFT) latency by 42% via custom connection pooling and pipeline optimization
- Owned system design discussions and cross-functional delivery, architecting backend services from scratch that accelerated feature delivery cycles by 35%

Full Stack Developer Intern — Future Interns

Jun 2025 – Jul 2025

Remote

- Architected high-availability RESTful APIs for a headless backend with strict schema validation in Node.js and MongoDB, maintaining 99.9% reliability under peak load

PROJECTS

Orbit — End-to-end encrypted, horizontally scalable real-time platform

GitHub | Live Demo

React • Node.js/Express • Socket.IO • MongoDB • Redis • X3DH • AES-256-GCM

- Implemented client-side end-to-end encryption using X3DH session establishment, Double Ratchet key evolution, and AES-256-GCM message encryption, with ratchet state persisted locally through IndexedDB-backed session storage.
- Built passwordless Constellation authentication with Argon2id hashing, server-side peppering, single-use Redis-backed challenge nonces, CSRF protection, and progressive logout controls for replay and brute-force resistance.
- Scaled real-time messaging across stateless Node.js instances using Socket.IO rooms and the Redis adapter, adding idempotent message delivery, offline-message queuing, presence synchronization, and device-session revocation.

Forge AI — Full-stack SaaS platform with AI generation, persistence, and payments

GitHub | Live Demo

React • TypeScript • Node.js/Express • PostgreSQL/Neon • Prisma • Redis/BullMQ • Stripe

- Developed a React 19 and TypeScript SaaS frontend with Vite, Tailwind CSS, React Router, GSAP, and reusable components for prompt-driven website generation, live previews, project management, deployments, monitoring, authentication, and version navigation.
- Designed Express 5 REST and WebSocket services integrating Better Auth, Prisma-backed Neon PostgreSQL, Redis/BullMQ workflows, and Qwen2.5-Coder model routing for initial generation and iterative code refinement.
- Implemented credit-based usage accounting with Stripe webhook-signature verification, transaction records, payment-state checks, and database-backed credit allocation after successful payment confirmation.

SENTINEL-AI — Low-latency semantic retrieval & automated support triage

GitHub | Live Demo

Python • FastAPI/Starlette • FAISS • Sentence Transformers • Pydantic v2 • Gradio • Groq

- Built a two-pass support-triage pipeline combining Pydantic-validated intent classification, calibrated escalation gating, in-memory FAISS IndexFlatIP retrieval, and policy-constrained response drafting over 15,000 historical resolutions.
- Reduced semantic vector lookup from an approximately 1.5-second remote-vector-store baseline to under 1ms using normalized local embeddings and an in-memory FAISS index, while maintaining deterministic temperature-zero response generation.
- Created an offline-reproducible evaluation harness with disk-backed caches and a 250-case human-annotated golden set, achieving a 34.9% escalation false-negative rate, 0.583 intent Macro-F1, and 0.558 Cohen's Kappa.

TECHNICAL SKILLS

- Languages:** Java, C++, Dart, Kotlin
- Backend & APIs:** Django, Mongoose, GraphQL
- Databases & Infrastructure:** MySQL, SQLite, AWS, Kubernetes
- Security:** JWT, WebAuthn, HMAC-SHA256
- Testing & Tools:** Git, GitHub Actions, Jest, Supertest

ACHIEVEMENTS & CERTIFICATIONS

- Top 15 Finalist**, Microsoft MSC Hackathon – selected out of 250+ competing teams
- Participant**, HackSRM, SRM University
- Oracle Certified** – Java SE 17 Developer (2026) | **MongoDB Certified** – Associate Developer (2026)

EDUCATION

Bachelor of Technology in Computer Science & Engineering (AI/ML Specialization)

Aug 2023 – May 2027

SRM University, Andhra Pradesh

Amaravati, India

Relevant Coursework: Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks, Distributed Systems, Machine Learning